

Training-Induced Changes in Neural Function

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AAGAARD, P. Training-induced changes in neural functions. *Exerc. Sport Sci. Rev.*, Vol. 31, No. 2, pp. 61–67, 2003. *Adaptive changes can occur in the nervous system in response to training. Electromyography studies have indicated adaptation mechanisms that may contribute to an increased efferent neuronal outflow with training, including increases in maximal firing frequency, increased excitability and decreased presynaptic inhibition of spinal motor neurons, and downregulation of inhibitory pathways.* **Keywords:** CNS, spinal, human, motor neurons, muscle

INTRODUCTION

Adaptive alterations can be induced in the neuromuscular system in response to specific types of training. Thus, increases in maximal contraction force and power as well as maximal rate of force development (RFD) will occur not only because of alterations in muscle morphology and architecture (2), but also as a result of changes in the nervous system (1,4,12).

Evidence of the adaptive change in neural function with training has been provided through the use of electromyography (EMG). Although consistent data can be obtained by EMG recording (7), inherent methodological constraints may sometimes exist for the measurement of surface EMG during voluntary muscle contraction. To overcome some of these problems, EMG normalization procedures, single motor unit recording techniques, and measurements of evoked reflex responses (Hoffmann reflex, V-wave) have been increasingly used to examine the change in neural function induced by training.

CHANGES IN EFFERENT NEURAL DRIVE ASSESSED BY EMG

The interference EMG (Fig. 1) comprises the composite sum of all the muscle fiber action potentials present within

the pickup volume of the recording electrodes. This overall interference signal is modified by a multitude of intracellular and extracellular factors, which all exert a significant influence on the pattern of spatial and temporal summation of the single action potentials. From a physiological perspective, the EMG interference signal is a complex outcome of motor unit recruitment and firing frequency (rate coding) that also reflects changes in the net summation pattern of motor unit potentials, as occurs with motor unit synchronization. Numerous studies have reported increased EMG amplitude after resistance training. The training-induced increase in EMG that has been observed in highly trained strength athletes indicates that neural plasticity also exists in subjects with highly optimized neural function.

Substantial cancellation of the EMG interference signal can occur due to out-of-phase summation of motor unit action potentials (MUAPs), and it has been suggested, therefore, that the EMG interference amplitude does not provide a true estimate of the total amount of motor unit activity (6). For example, increased motor unit synchronization will cause the EMG signal amplitude to increase (16) attributable to the elevated incidence of in-phase MUAP summation. Consequently, the increase in EMG interference amplitude observed after resistance training could indicate changes in motor unit recruitment, firing frequency, and MUAP synchronization. In addition, not all studies have been able to demonstrate elevated EMG activity after resistance training. The inability to detect longitudinal EMG changes could be, at least in part, due to changes in skin and muscle tissue properties (subcutaneous fat layer, muscle fiber pennation angle) or could arise from changes in electrode positions between testing sessions. As discussed below, however, it is possible to reduce some of these limitations by employing intramuscular EMG recordings, EMG normalization procedures, and measurements of evoked reflex responses.

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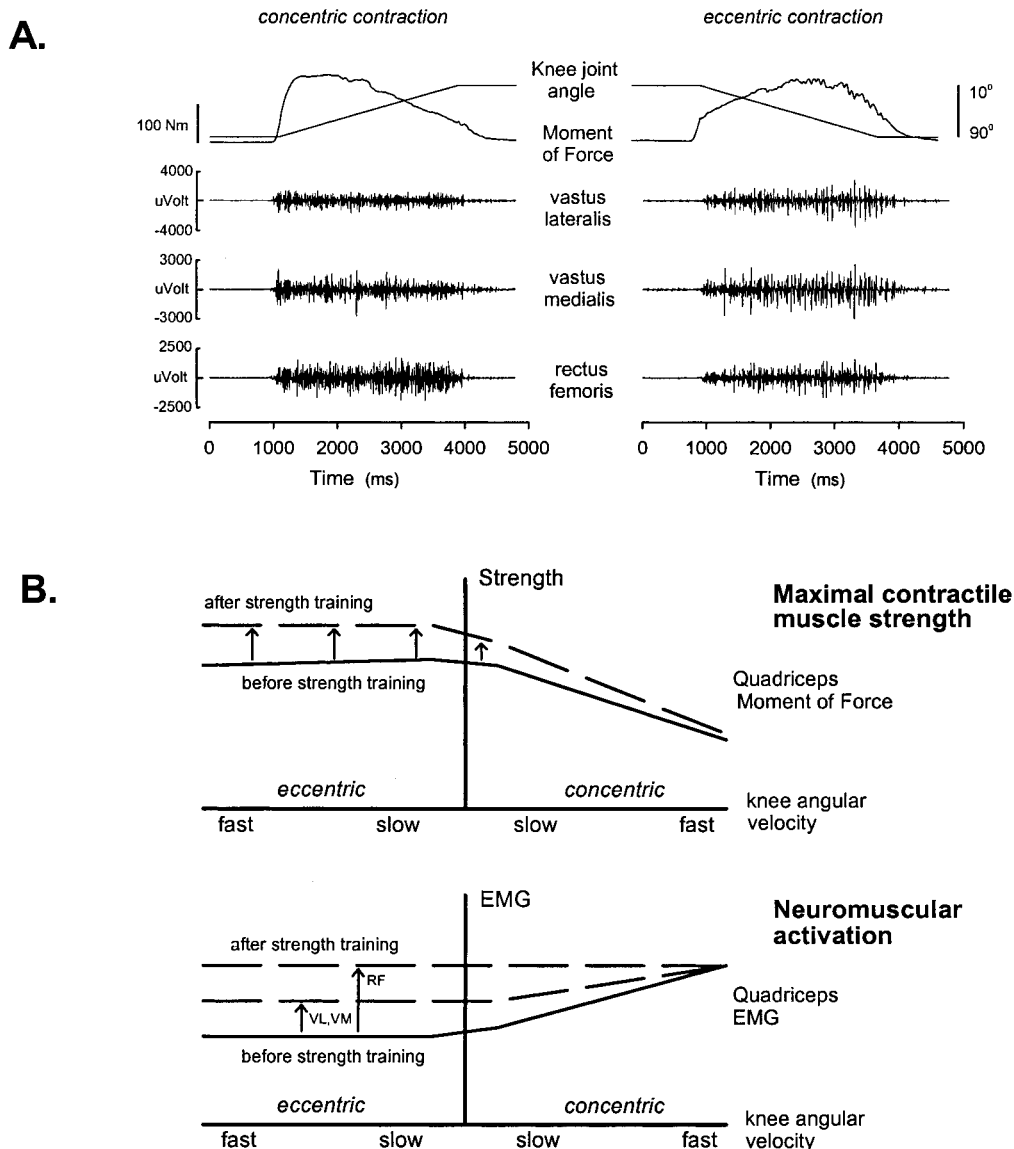


Figure 1. A. Knee joint moment (Moment of Force) and interference EMG recorded in an untrained subject during maximal effort concentric and eccentric contractions of the quadriceps muscle performed in an isokinetic dynamometer (knee joint angular velocity 30°s^{-1} , 90° – 10° = range of motion, 0° = full extension). During eccentric contraction large EMG spikes typically were observed separated by interspike periods of low or absent activity. This pattern was less frequent after intense resistance training, where the concentric and eccentric EMG signals became more similar. When rectified and low-pass filtered EMG signals were analyzed in untrained subjects, EMG amplitudes were 20–40% less during maximal eccentric than concentric contraction (see B). B. Resistance training with heavy loads has consistently been shown to increase maximal eccentric and slow concentric contraction strength (quadriceps femoris muscle, *top curve*). In these specific contraction conditions, which are characterized by high levels of contractile force generation, muscle activation appears to be suppressed in untrained subjects despite a maximal voluntary effort (EMG, *bottom curve*). After a regimen of prolonged resistance training with heavy loads, the suppression of the EMG signal amplitudes was fully abolished (rectus femoris (RF)) or partially removed (lateral vasti (VL) and medial vasti (VM)) in parallel with a marked increase in maximal eccentric muscle strength. Data from (1).

CHANGES IN MOTONEURON FIRING FREQUENCY

Motor unit firing rates have been recorded at much higher frequencies than that needed to achieve full tetanic fusion in force. For example, firing rates of 100–200 Hz can be observed at the onset of maximal voluntary muscle contraction (12), with much lower rates (15–35 Hz) at the instant of maximal force generation (MVC), which typically occurs 250–400 ms after the onset of contraction. Importantly, firing frequency has a strong influence on the contractile rate of force development. In fact, the rate of force development

continues to increase at stimulation rates higher than that needed to achieve maximum tetanic tension (Fig. 2) (9). It is possible, therefore, that supramaximal firing rates in the initial phase of a muscle contraction serve to maximize the rate of force development rather than to influence maximal contraction force *per se*. When contractile force is less than the maximal tetanized level, it can be temporarily elevated by the addition of an extra discharge pulse (1–5 ms interpulse interval), as demonstrated using constant-frequency stimulation of single motor units, whole isolated muscle, and intact human muscle. This phenomenon has been referred to as the

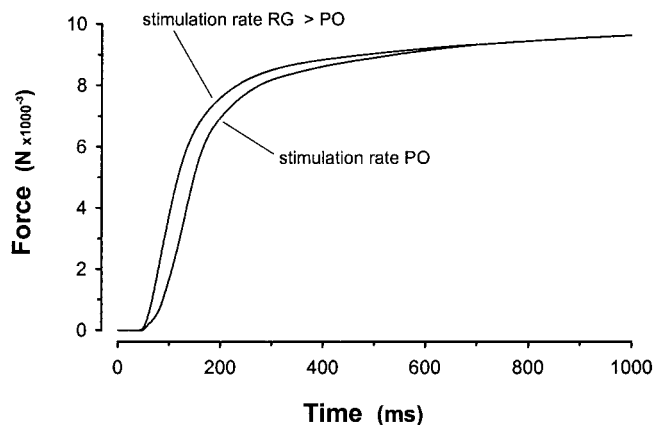


Figure 2. Force-time curves for isolated motor units in the rat soleus muscle when activated at the minimum frequency needed to elicit maximal tetanic fusion (PO), and when activated at a supramaximal rate (RG) that also elicited maximal tetanic fusion. Note that the rate of force development is greater at supramaximal rate of stimulation. [Adapted from Nelson, A.G. Supramaximal activation increases motor unit velocity of unloaded shortening. *J. Appl. Biomech.* 12:285–291, 1996. Copyright © 1996 Human Kinetics Publishers. Used with permission.]

catch-like property of skeletal muscle. At the onset of rapid muscle contractions *in vivo*, so-called discharge doublets (interspike interval < 10 ms) may be observed in the firing pattern of single motor neurons (see (12)). Although the functional consequences of such discharge doublets are not fully understood, it is possible that the firing of discharge doublets at the onset of contraction and during the phase of rising muscle force serves to enhance the initial generation of muscle contraction force by taking advantage of the catch-like property, hence increasing the rate of force development. Interestingly, ballistic-type resistance training, *i.e.*, involving maximal intentional rate of force development, markedly increased the incidence of discharge doublets in the firing pattern of individual motor units (from 5% to 33%) while also increasing the rate of force development (12).

The maximal firing frequency of motor units can be examined by use of intramuscular EMG-recording techniques (multipolar needle, wire electrodes). Based on such techniques, the frequency of muscle fiber action potentials obtained during maximal voluntary contraction was significantly greater in trained elderly weight lifters compared with age-matched untrained individuals. Moreover, maximal firing frequency has been reported to increase in response to resistance training (Fig. 3). Training-induced increases in the maximal frequency of muscle fiber action potentials appear to occur in both young and elderly individuals. Although elderly subjects demonstrate a lower maximal discharge rate than young subjects, this difference is reduced with resistance training. Thus, the increase in maximal firing frequency induced by resistance training may effectively overrule the age-related decline in maximal discharge rate. This would represent a highly beneficial type of neural adaptation to counteract the gradual decline in activation of muscle fibers and the associated impairment in muscle function observed with increasing age.

CHANGES IN RATE OF FORCE RISE AND EMG DEVELOPMENT

An increase in the rate of force development is perhaps the single most important functional benefit induced by resistance training. Rapid movements may involve muscle contraction times of 50 to 200 ms, which are considerably less than the time it takes to reach maximal muscle force (~300 ms). A training-induced increase in the rate of force development, therefore, makes it possible to reach a higher force and velocity during fast movements. Importantly, the rate of force development plays an important role in the ability to perform rapid and forceful movements, both in highly trained athletes as well as elderly individuals who need to control unexpected perturbations in postural balance.

Acutely, the rate of force development is enhanced with an increase in efferent neural drive, particularly by increases in the firing frequency of motor units (Fig. 2). Parallel increases in the rate of force development and EMG amplitude have been observed after resistance training (3,12). In particular, a marked increase in EMG amplitude and the rate of rise in EMG can be seen in the initial contraction phase (Fig. 4), which suggests that neural adaptation mechanisms, including an elevated incidence of discharge doublets (12), are highly important for the training-induced increase in the rate of force development. As described above, Duchateau and colleagues recently reported concurrent increases in the rate of force development and maximal firing frequency, together with a sixfold increase in the incidence of discharge doublets in the firing pattern of individual motor units following resistance training (12). This elevated incidence in discharge doublets and the corresponding rise in initial motoneuron firing frequency probably represent major mechanisms re-

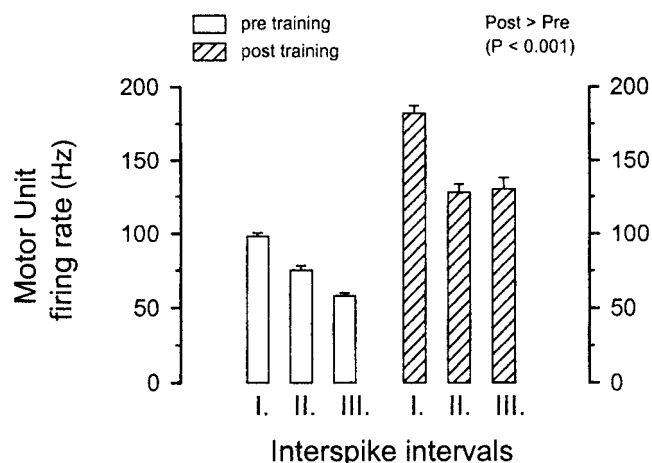


Figure 3. Instantaneous motor unit firing frequency ($\pm$ SEM) at the onset of maximal ballistic contractions, recorded in the tibialis anterior muscle before and after a period of ballistic-type resistance training, *i.e.*, performed at maximal intentional rate of force development. Bars show the mean discharge frequency recorded in the initial, second, and third time intervals between successive action potentials. An increase in motoneuron firing frequency was observed following training, as all post-training values were greater than pretraining values ($P < 0.001$). The number of discharge signals analyzed in each interspike period ranged between 243 and 609. Increases in firing frequency appeared to occur independently of motor unit size, as changes were not related to either time to peak tension or the recruitment threshold. Data from (12).

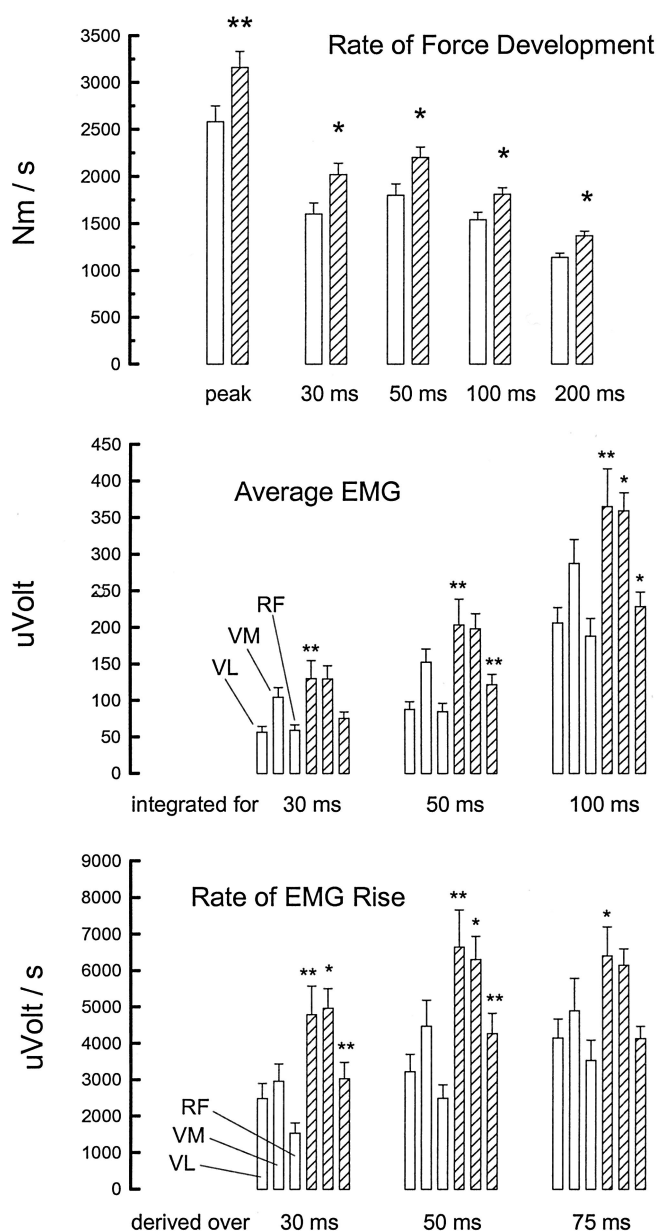


Figure 4. Contractile rate of force development (RFD) and EMG (average EMG and rate of EMG rise) obtained in the quadriceps femoris muscle (vastus lateralis (VL), vastus medialis (VM), rectus femoris (RF)) during maximal isometric contraction before (open bars) and after (closed bars) 14 wk of resistance training. Time intervals denote time relative to contraction onset (for RFD) or onset of EMG (for all EMG parameters). Post > pre: RFD and average EMG. * $P < 0.05$; ** $P < 0.01$, rate of EMG rise; * $P < 0.01$; ** $P < 0.001$. Data from (3).

sponsible for the increase in the rate of force development observed with training. Furthermore, training-induced changes in muscle fiber size and muscle architecture (2) would additionally contribute to the increase in rate of force development.

The enhancement of motor unit activity may involve changes in neural circuitry. Recurrent Renshaw inhibition of spinal motor neurons, for example, has been considered as a limiting factor for discharge rate, and has been considered to have a regulating influence on the reciprocal Ia-inhibitory pathway. Animal experiments have shown that Renshaw

cells receive several types of supraspinal synaptic input that can enhance as well as depress the recurrent pathway. Compared with steady-force contractions, Renshaw cell activity appears to be more inhibited during maximal phasic muscle contractions, which results in reduced recurrent inhibition. This suggests that explosive-type resistance training (*i.e.*, training involving a high rate of force development) may be optimal for evoking changes in maximal firing rate of motor units. However, this effect may be restricted to certain muscles, as recurrent inhibition appears to be absent in the smaller distal muscles of the hands and feet.

CHANGES IN NEURAL INNERVATION DURING MAXIMAL ECCENTRIC MUSCLE CONTRACTION

It has been suggested that eccentric muscle contractions require unique neural activation strategies. Thus, preferential activation of high-threshold motor units has been observed in the triceps surae muscle during submaximal eccentric contraction, which was suggested to result from increased presynaptic inhibition of Ia afferents that synapse onto low-threshold motor neurons. However, the majority of studies, using single motor unit recordings in muscles of the hand and lower back, have failed to demonstrate selective recruitment of high-threshold motor units during eccentric contractions. Interestingly, the excitability of spinal motor neurons or presynaptic inhibition of Ia afferents in the soleus muscle appears to differ between submaximal eccentric and concentric contractions at matched EMG levels, as suggested by a depression in H-reflex amplitude during eccentric contraction.

Electrical stimulation of passive versus active muscle has been used to address the issue of neural activation during maximal eccentric contractions. Despite a maximal intended effort by the subjects, the force achieved during maximal eccentric contraction was enhanced with superimposed electrical stimulation, whereas no effect was observed during concentric contractions (14). Notably, this evoked increase in eccentric contraction strength was seen in untrained subjects but not in strength-trained subjects, suggesting that the apparent inhibition in maximal eccentric muscle strength can be removed by resistance training.

Electromyography recordings in untrained subjects have shown that muscle activation is suppressed during maximal eccentric contractions (Fig. 1), as EMG is reduced compared with maximal concentric contraction (1). Importantly, this inhibition in muscle activation appears to be downregulated or fully removed in response to resistance training with heavy loads (1), which explains the marked increase in maximal eccentric strength typically observed with this type of training.

Although several mechanisms have been proposed, the actual neural regulatory pathways responsible for the suppression of muscle activation during eccentric contraction remain unidentified. Efferent motor output during maximal voluntary muscle contraction not only is regulated by central descending pathways, but also is modulated by afferent inflow from group Ib Golgi organ afferents, group Ia and II muscle spindle afferents, group III muscle affer-

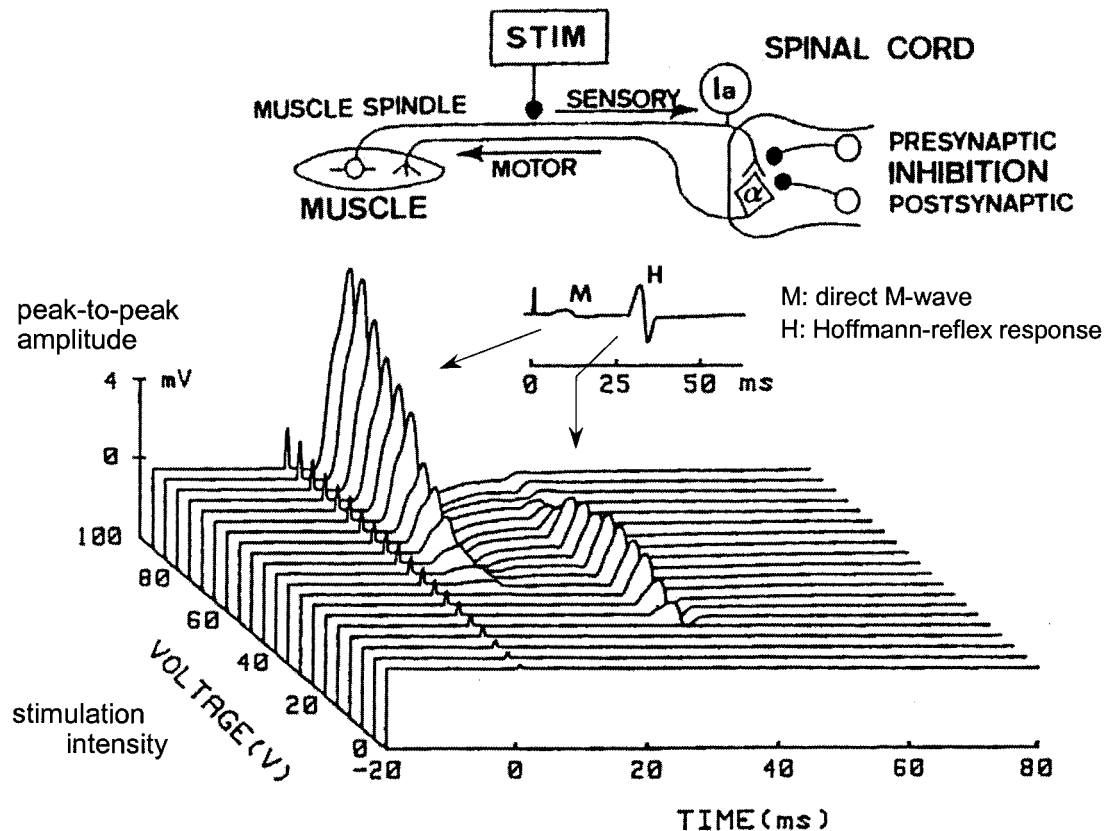


Figure 5. Evoked spinal motoneuron responses examined by use of the Hoffmann (H) reflex. Electrical stimulation of Ia afferent nerve fibers ("sensory") excites spinal motor neurons (α), in turn eliciting a reflex response (H) with a latency of 30–35 ms for the human soleus muscle. The electrical pulse also gives rise to action potentials in the motoneuron axones ("motor") resulting in a direct M-wave with a latency of 3–4 ms. A variant of the H-reflex, so-called V-waves, can be recorded using supramaximal stimulation intensities during ongoing maximal muscle contraction. Elevated V-wave and H-reflex responses have been observed following resistance training, indicating an elevated descending motor drive from supraspinal centers, increased excitability of spinal motor neurons and/or decreased presynaptic inhibition of muscle spindle Ia afferents. [Adapted from Moritani, T., and Y. Yoshitake. The use of electromyography in applied physiology. *J. Electromyogr. Kinesiol.* 8:363–381, 1998. Copyright © 1998 Elsevier Science; and Stein, R.B., and C. Capaday. The modulation of human reflexes during functional motor tasks. *Trends Neurosci.* 11:328–332, 1988. Copyright © 1988 Elsevier Applied Science Publishing. Used with permission.]

ents, and by recurrent inhibition from Renshaw cells. All of these pathways are expected to exhibit adaptive plasticity with training (5). For example, Golgi Ib afferents activate interneurons in the spinal cord, which are also influenced by descending corticospinal pathways. It has been suggested that the removal of neural inhibition and the corresponding increase in maximal eccentric muscle strength observed following resistance training could be caused by a downregulation of such spinal inhibitory interneuron activity, possibly by central descending pathways (1). Not only have reduced H-reflex responses been observed during active eccentric versus concentric contractions (see above), the H-reflex also appears to be markedly suppressed during passive lengthening compared with shortening of the soleus muscle. The possibility exists, therefore, that in eccentric contraction the spinal inflow from Golgi Ib afferents and joint afferents induce elevated presynaptic inhibition of muscle spindle Ia afferents, thereby reducing the magnitude of excitatory inflow to motor neurons. Consequently, a training-induced reduction in presynaptic inhibition of Ia muscle spindle afferents could contribute to an elevated excitatory inflow

to spinal motor neurons during maximal eccentric muscle contraction.

Future studies are needed to address the training-induced change in excitability and postsynaptic inhibition of spinal motor neurons as well as the possible alteration in presynaptic Ia afferent inhibition during maximal eccentric contraction. Also, additional studies using intramuscular EMG techniques are needed to examine further whether the specific recruitment order and firing rate of individual motor units in fact differ between eccentric and concentric muscle contraction of both submaximal and maximal intensity, and if such difference is muscle specific.

CHANGES IN EVOKED SPINAL MOTONEURON RESPONSES

Few studies have measured responses in spinal motoneuron to examine the importance of neural mechanisms for the training-induced increase in maximal muscle strength. The Hoffmann (H) reflex may be useful for the assessment of motoneuron excitability *in vivo*, although it also reflects the

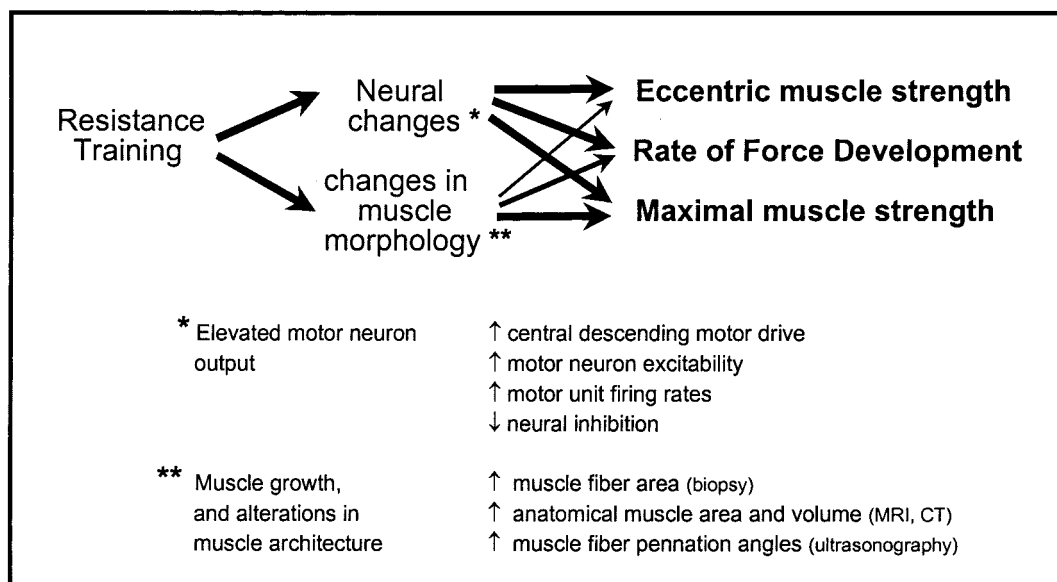


Figure 6. Resistance training can induce adaptive alterations in nervous system function, along with changes in the morphology and architecture of the trained muscles. In particular, neural adaptation mechanisms play important roles for the training-induced increase in maximal eccentric strength and contractile rate of force development (RFD). *Thick arrows* indicate a strong influence, *thinner arrows* a moderate influence, and *thinnest arrows* indicate a low-to-moderate influence. Resistance training aimed at maximizing neural components will induce gains in muscle strength with no or only minor increases in muscle and body mass, which will benefit certain individuals and athletes (*i.e.*, distance runners, triathletes, cyclists). Training that results in both improved neural function and gains in muscle mass will benefit not only explosive-type athletes but also aged individuals, as for the frail elderly this will provide an effective mean to improve everyday physical function.

magnitude of presynaptic inhibition of Ia afferent synapses (see (15)). In brief, the H-reflex is elicited by electrical stimulation of the peripheral nerve containing Ia afferents and motor axons (Fig. 5) (see (4)). In the resting muscle, the amplitude of the H-reflex varies nonmonotonically with stimulus intensity, achieving maximum amplitude at an intermediate stimulus intensity. A supramaximal intensity elicits a maximal direct M response (M_{max}) by orthodromically activating all motor axons in the peripheral nerve. At high intensities of stimulation, the H-reflex response is abolished due to collision between (i) action potentials that travel antidromically in the motor axon toward the spinal cord and (ii) action potentials that propagate orthodromically from the spinal cord toward the muscle fibers due to the volley of H-reflex impulses. When supramaximal nerve stimulation is superimposed during ongoing voluntary muscle contraction, the H-reflex response reappears (denoted a V-wave) as the antidromic impulses in the motor axons now collide with efferent nerve impulses generated by the voluntary motor effort (11). An increased central descending motor drive results in an increased motor neuron recruitment and firing rate, which increases the outflow of efferent motor impulses in the axons. Hence, any increase in descending motor drive will produce an increased cancellation of the antidromic impulses, thus allowing more of the evoked H-reflex volley to reach the muscle fibers as manifested by an increase in V-wave amplitude (4,11). Likewise, an increased excitability of spinal motor neurons or reduced presynaptic inhibition of Ia afferents would contribute to the observed increase in V-wave amplitude. It has been observed, however, that presynaptic inhibition of soleus Ia afferents (from reciprocal inhibitory pathways) is absent or greatly reduced during maximal plantarflexor contraction in young subjects. It should

also be recognized that training-induced changes in postsynaptic inhibition, such as via Golgi Ib afferents, could contribute to an increase in evoked V-wave and H-reflex responses.

In cross-sectional studies, elite weight lifters and sprinters have demonstrated markedly elevated V-wave amplitudes in the hand and lower limb muscles compared with untrained subjects, which was interpreted as an increased ability to activate motor units during maximal voluntary contraction. Furthermore, resistance training appears to induce increased V-wave and H-reflex amplitudes during maximal muscle contraction (4). It is difficult to elicit V-wave changes in certain hand muscles with resistance training, suggesting that the relative contribution of the above mechanisms may differ between various muscles in the body.

Considerable training-induced plasticity appears to exist for the excitatory and inhibitory pathways in the spinal cord. Upregulation and downregulation of the H-reflex have both been demonstrated in monkeys and rats exposed to long-term conditioning paradigms. Excitability of the soleus H-reflex decreased after short-term balance training by human subjects, which likely reflects altered states in the reciprocal inhibitory pathways of the tibialis and soleus muscles. From a functional perspective, it would seem desirable to achieve a suppressed stretch-reflex response in antagonist-agonist muscle pairs during postural balance tasks, to avoid the occurrence of reflex-mediated joint oscillation. Elevated H-reflex responses have been obtained during submaximal and maximal voluntary muscle contraction after hopping training (13) and resistance training (4), respectively. Notably, the H-reflex response recorded at rest remained unchanged with training (4,13). These findings indicate that resting H-reflex measurements may not adequately reflect the state of the

spinal circuitry during activity, advocating that evoked reflex measurements should be performed in functional contraction tasks and not solely in the resting muscle. Also of importance, normalizing the evoked H-reflex and V-wave responses to the maximal M-wave minimizes the methodological limitations associated with conventional surface EMG recordings.

CONCLUSIONS

Resistance training elicits adaptive changes in the nervous system as well as in the morphology of the trained muscles (Fig. 6). In particular, neural adaptation mechanisms are important for the increases in maximal eccentric strength and rate of force development observed with training. The increase in motor neuronal output in response to resistance training may involve increased firing rates, increased motoneuron excitability and decreased presynaptic inhibition, downregulation of inhibitory neural pathways, as well as increased levels of central descending motor drive. Further research is needed to obtain knowledge about the relative involvement and functional significance of these neural factors with specific types of physical activity and training, and to examine the relative importance of these mechanisms in young versus elderly individuals.

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